

TOPIC:-

Activity 1 Collect photographs of 5 plants each of the Thallophyta, Bryophyte and Pteridophyta divisions and write a description of each.

Introduction:-

Thallophyta, Bryophyta, and Pteridophyta are flowerless, seedless lower plants. They are also called cryptogams, which are plants that reproduce by spores without flowers or seeds.

- Thallophyta:- Thallophyta refers to seedless and flowerless organisms consisting of algae, fungi, and lichens.

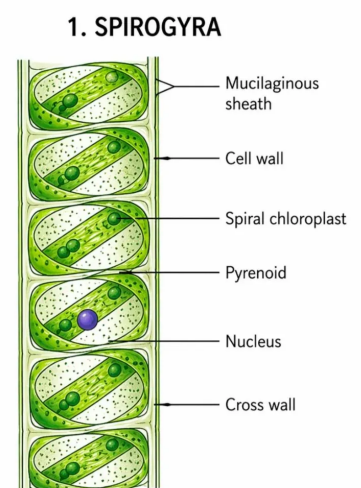
- Bryophyta :- Bryophyta refers to small, flowerless plants consisting of mosses and liverworts.

- Pteridophyta :-Pteridophyta refers to flowerless plants consisting of ferns and their relatives.

Further, we will understand it with deeper description and examples of each.

◆ Thallophyta Division :-

1. Spirogyra:-



Spirogyra are free floating green algae present in freshwater bodies. They belong to the division Thallophyta. Spirogyra are commonly known as water silk or pond silk.

There are around 400 species of Spirogyra. They increase the level of oxygen in their habitat.

Classification of Spirogyra:-

Spirogyra is classified under Chlorophyceae due to the presence of chlorophyll.

The genus contains around 400 species.

◎Structure of Spirogyra:-

✿It grows mainly in water.

✿It does not have specific parts like roots, stem, leaves, or flowers but are autotrophic due to the presence of chlorophyll.

✿They have a soft and thread like body.

✿It has spirally arranged chloroplasts in its cells.

✿The vegetative structure of Spirogyra is an unbranched filamentous thallus.

Spirogyra Details and Reproduction

✿They are 10 to 100 micrometre in width and may grow several centimetres in length.

✿They are present as a slimy mass due to the presence of mucilage sheath around the filament.

✿The cell wall is made up of two layers inner cellulose and outer pectose.

☉Spirogyra Reproduction:-

✿Spirogyra can reproduce both sexually and asexually.

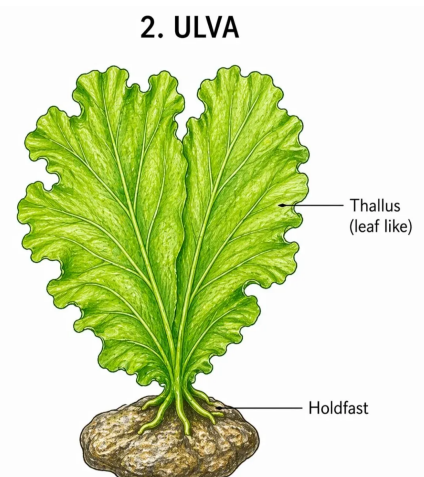
✿Asexual or vegetative reproduction occurs by simple fragmentation of the filaments.

✿Sexual reproduction occurs by a process known as conjugation, in which cells of two filaments lying side by side are joined by outgrowths called conjugation tubes.

✿This allows the contents of one cell to pass into the other.

✿The resulting cell called zygote becomes surrounded by a thick wall and overwinters while the vegetative filaments die.

2. ULVA:-



Ulva belongs to the Ulvaceae family. *Ulva intestinalis* is a green algae also known as gutweed or sea lettuce. It grows in marine environments. Earlier, classification was based on morphology, but later it was reclassified using genetic work completed in the early 2000s.

Classification of Ulva:-

Ulva classified under the division of chlorophyta .It is a genus of thallophyta they are more than 125 pieces of ulva currently accepted taxonomically worldwide.

Kingdom: Plantae

Division: Chlorophyta

Class: Ulvophyceae

Order: Ulvales

Family: Ulvaceae

Genus: Ulva

☉Structure of Ulva:-

✿Ulva does not have true roots, stems, leaves or flowers.

✿They are autotrophic due to the presence of chlorophyll.

✿Ulva are multicellular organisms.

✿They have a soft and leaf-like body.

✿Ulva can grow up to 30 to 50 cm in length but may also remain smaller.

✿The thallus is expanded and consists of two layers of cells. It develops from a single cell by cell division.

✿The base grows into a short stalk which later forms a holdfast that attaches the algae to the substrate.

✿Cell division can occur in any part of the thallus.

☉Ulva Reproduction:-

✿Ulva can reproduce both sexually and asexually.

✿Vegetative reproduction takes place by fragmentation of the thallus, usually in calm water.

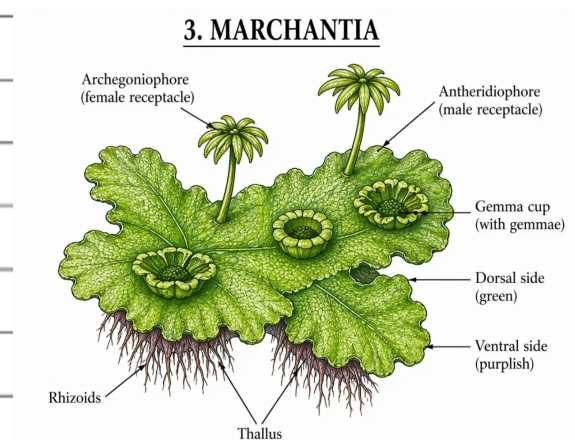
✿ Asexual reproduction in *Ulva* occurs through quadriflagellate zoospores formed inside vegetative cells.

✿ Sexual reproduction is usually isogamous. Gametes are formed in the cells of the haploid thallus.

✿ *Ulva* shows isomorphic alternation of generations, where both generations look similar.

◆ BRYOPHYTA DIVISION:-

1. Marchantia:-



Marchantia is a genus of liverworts. It is a small, green, ribbon-like plant that grows in moist and shady places. It is commonly found on moist soil, especially in recently burned land and throughout the Northern Hemisphere. *Marchantia* belongs to the division Bryophyta. There are more than 9000 species of liverworts on earth.

Classification of *Marchantia*:-

Marchantia is classified under the division Hepaticophyta, which includes all the liverworts. Bryophyta is divided into three main classes:

Hepaticopsida - contains liverworts

Anthocerotopsida - contains hornworts

Bryopsida - contains mosses

Kingdom: Plantae

Division: Hepaticophyta

Class: Hepaticopsida

Order: Marchantiales

Family: Marchantiaceae

Genus: Marchantia

☉Structure of Marchantia:-

✿Marchantia is called an amphibious plant because it grows in moist soil and requires water for reproduction.

✿It is thalloid, multicellular and autotrophic.

✿The plant body is ribbon-like and does not have true roots, stems or leaves.

Instead, it has leaf-like structures and root-like rhizoids.

✿It lacks vascular tissues for transport of water and food.

✿It commonly grows in moist and shady habitats.

✿The thallus is dorsiventral, flat and dichotomously branched.

✿The gametophyte is the dominant stage of the life cycle.

✿The upper surface has pores for gaseous exchange.

☉Marchantia Reproduction:-

✿Marchantia reproduces both sexually and asexually.

✿Sexual reproduction in Marchantia is by fragmentation or by forming specialized structures known as gemmae.

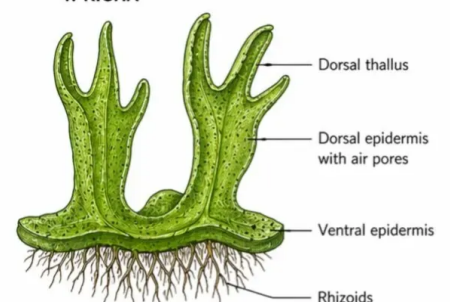
Marchantia is dioecious.

✿Male and female sex organs are antheridia and the female reproductive organ is archegonia.

✿They bear on the mature gametophyte and are erect and modified stalks known as antheridiophore and archegoniophore respectively.

2.RICCIA:-

4. RICHA



Riccia genus of liverworts (small, creeping plants) in order Marchantiales, widely distributed throughout the world. The most well-known species *Riccia fluitans*, sometimes called slender riccia, forms branching green ribbons about 0.1 centimetre (about 0.04 inch) wide and about 1.3 to 5 centimetres long that float in shallow ponds. The ribbons often become tangled in large masses. Other species of *Riccia* form rosettes on moist soils.

Classification of Riccia:-

Riccia is classified under division Marchantiophyta which includes all the liverworts.

Kingdom: Plantae

Division: Marchantiophyta

Class: Marchantiopsida

Order: Marchantiales

Family: Ricciaceae

Genus: Riccia

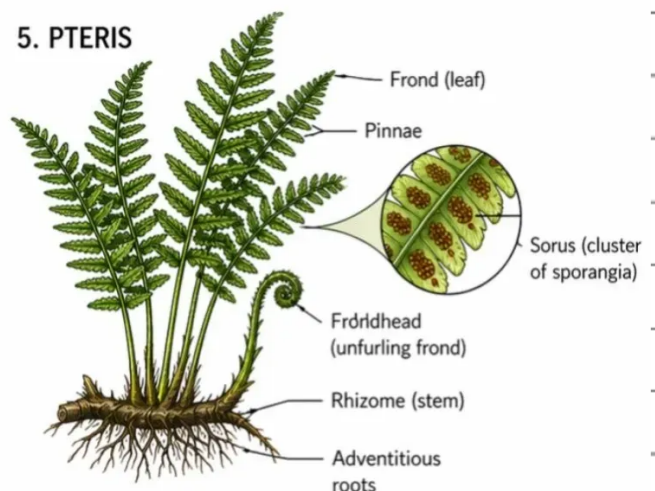
☉ Structure of Riccia:-

✿ The plant body of *Riccia* is gametophytic.

✿ The gametophyte is a flat, dichotomously branched, green, dorsiventral, prostrate, fleshy thallus.

◆ Pteridophyta division:-

PTERIS:-



Pteris vittata, commonly known variously as the Chinese brake, Chinese ladder brake, or simply ladder brake, is a fern species in the Pteridaceae subfamily of the Pteridoideae. It is indigenous to Asia, southern Europe, tropical Africa and Australasia. The type specimen was collected in China by Pehr Osbeck.

Classification of *Pteris*

Pteris is classified under the division of pteridophyta. There are about 300 species of ferns in the subfamily Pteridoideae of family Pteridaceae.

Kingdom: Plantae

Division: Pteridophyta

Class: Polypodiopsida

Order: Polypodiales

Family: Pteridaceae

Genus: *Pteris*

☉ Structure of *Pteris*

- ✿ *Pteris* have well-developed roots, stem and leaves.

- ✿ They do not bear flowers and fruits.

- ✿ They have well-developed tissues for conduction of food and water.

- ✿ Each branch of the thallus is linear to wedge-shaped or obcordate.

- ✿ The thallus is 5–8 mm long and 1–2 mm wide in larger species, while in *Riccia* it is 30–50 mm long and 1–2 mm broad.

- ✿ The ventral surface of *Riccia* thallus bears numerous scales and rhizoids.

- ✿ The rhizoids are unicellular, elongated, unbranched tubular hair-like structures present on the ventral surface of the thallus.

☉ *Riccia* Reproduction:-

- ✿ *Riccia* can reproduce sexually and asexually both.

- ✿ Asexual reproduction in *Riccia* is by vegetative reproduction.

- ✿ The male gametes and the female gametes are structurally and physiologically different.

■REFERENCES:-

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